Dataset Datasheet

Course: DSPRO2 HS26

Project Name:	Football Analyzer
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GitHub Repository:	`github.com/LittleBigFatFish/dspro2-sports-analyzer`

We use six third-party football datasets, including access-controlled SoccerNet material. Five contain images and are used for player and ball detection; SkillCorner contains tracking data for the play analysis. Every field below covers all six.

Dataset Overview

- **Official Name, Owner, Version & Date, Short Description, Reference:** see Table 1.

Official name	Owner	Version, date	Short description	Reference
Football Image Dataset	skaczylo (GitHub)	Commit `2642128`, 2026-06-23; downloaded 2026-10-20	612 football frames with ball, player and referee boxes in YOLO format.	`github.com/skaczylo/Football-Image-Dataset`
Soccana V1 (`Soccana_player_ball_detection_v1`)	Adit-jain (Hugging Face)	Revision `9e76891`, 2025-08-28; downloaded 2026-11-03	Player, referee and ball boxes merged from Roboflow Universe sets; mostly image tiles.	`huggingface.co/datasets/Adit-jain/Soccana_player_ball_detection_v1`
SoccerNet Tracking 2023	SoccerNet team (University of Liège, KAUST)	Revision `3777f45`, 2026-09-06; downloaded 2026-11-03	30 s broadcast clips with a tracked box per object, role, team and jersey.	A. Cioppa et al., CVPR Workshops 2022, arXiv:2204.06918
SoccerNet Game State Reconstruction 2025	SoccerNet team	Revision `ce84ee7`, 2024-10-22, labels v1.3; downloaded 2026-11-03	Broadcast clips with boxes, pitch positions, teams and jerseys for a minimap.	V. Somers et al., CVPR Workshops 2024, arXiv:2404.11335
SkillCorner Open Data	SkillCorner, with PySport	Commit `4340d27`, 2026-09-14; downloaded 2026-11-03	Tracking of players and ball at 10 fps and events for 20 A-League matches; no images.	`github.com/SkillCorner/opendata`
MultiSports (football part)	Multimedia Computing Group, Nanjing University	Revision `01600ce`, 2022-12-13; downloaded 2026-11-03	Broadcast clips with action tubes: one box per frame on the player doing the action.	Y. Li et al., ICCV 2021, arXiv:2105.07404

Table 1: The six datasets.

Motivation and Intended Use

- **Primary Purpose:** Football: a ready-to-use YOLO training set. Soccana: a starting point for a general football detector. SoccerNet Tracking and GSR: research benchmarks for tracking and minimap reconstruction. SkillCorner: open tracking data for analytics. MultiSports: a benchmark for action detection.
- **Intended Use:** Comparing pretrained YOLOv8n, YOLO12n and YOLO26n at 640 and 1280 px on all image datasets, then fine-tuning (Football, Soccana, SoccerNet train). SoccerNet test and GSR valid are the main evaluation sets. The counts below describe the pretrained-model comparison; after fine-tuning, training images must not be reported as independent tests. Overlapping sources must be excluded from the held-out evaluation. Telling the two teams apart: SoccerNet

and GSR give the team side of every player, SkillCorner the team of every tracked player. SkillCorner provides derived reference data for possession and passes, with missing observations and possible tracking errors.
- **Out-of-Scope Use:** Identifying people; team results on Football, Soccana or MultiSports (no team labels); claims beyond the leagues and games in the data; video-game frames reported as real footage; Soccana test or GSR test/train as separate results (they overlap); person precision or anything about the ball on MultiSports (only acting players are labelled); commercial use.

Composition and Coverage

- **Data Types:** Unstructured: JPEG images and frames (Football, Soccana, SoccerNet, GSR), MP4 videos (MultiSports). Labels as YOLO text, COCO JSON, MOT text, JSON and a pickle. SkillCorner is structured: JSON, JSON lines, CSV.
- **Population:** Professional football from TV broadcasts. Football: club and national-team matches, among them Bundesliga, LaLiga, Saudi league, J-League, Copa Libertadores, FA Cup and Community Shield, three highlight clips and one video game. Soccana: seven Roboflow source groups, 71% tiles. SoccerNet and GSR: 9 games, probably the Swiss Super League (boards visible, not documented). MultiSports: YouTube broadcasts, among them the London 2012 Olympics and English club games. SkillCorner: A-League 2024/25.
- **Dataset Size:** Five image datasets with 440,231 images and frames and 2,273,795 labelled boxes, of which 11,153 frames are used in our tests (Table 2).
- **Known Gaps / Imbalances:** The ball is tiny: median shorter side 5.0–5.6 px at 640 px input on SoccerNet and Football. SoccerNet and GSR cover only 9 games. 43 of 100 Football val images are from a video game. Soccana is mostly tiles with 3,501 empty labels. MultiSports labels only acting players and never the ball. SkillCorner has no tracking in 25.7% of in-play frames. Rain, snow, amateur and youth games are not established as covered by our audit.

Dataset	Images / frames	Boxes	Ball boxes	Used in tests
Football Image Dataset	612	10,136	500	612
Soccana V1	11,673	64,648	3,600	500
SoccerNet Tracking 2023 (106 clips)	79,500	1,297,549	75,060	3,180
SoccerNet GSR 2025, valid (58 clips)	43,500	748,666	40,900	1,740
MultiSports, football (534 clips)	304,946	152,796	–	5,121
Image datasets	**440,231**	**2,273,795**	**120,060**	**11,153**
SkillCorner (20 matches, tracking)	1,244,814	94,361 events	ball in 74%	–

Table 2: Size per dataset.

Sources and Collection

- **Provenance:** Third-party broadcast footage and video-game imagery, distributed through public or controlled-access sources. 48 of the 100 Football val images are also in Soccana train, so both share an upstream source. GSR test and train are byte-identical to SoccerNet Tracking. Our overlap check found no shared frames between MultiSports and Football or Soccana.
- **Collection Method:** Football: frames cut from videos. Soccana: source sets sliced into tiles and capped per source. SoccerNet and GSR: 30 s clips at 25 fps, 1920×1080. MultiSports: YouTube videos cut into clips, 1280×720 at 25 fps. SkillCorner: computer vision on broadcast video at 10 fps, positions outside the view extrapolated.
- **Jurisdictions:** Football: Germany, Spain, England, Saudi Arabia, Japan, South Korea, South America and international matches. SoccerNet and GSR: probably Switzerland. SkillCorner: Australia and New Zealand. Soccana and MultiSports: not documented.

Labels and Targets

- **Target Definition:** Boxes for ball, player (goalkeepers included) and referee, mapped to one shared scheme (0 ball, 1 player, 2 referee). SoccerNet "other" (staff) is ignored. Every MultiSports box counts as a player. For COCO models, player and referee count as "person", ball as "sports ball". Team labels: team side per player in SoccerNet and GSR, team per player in SkillCorner. SkillCorner: possessions and passes.
- **Labeling Process:** Football: manually annotated. Soccana: taken over from the source sets and merged. SoccerNet and GSR: by the authors (papers). MultiSports: professional athletes marked actions, crowd workers corrected the boxes. SkillCorner: events derived from the tracking data.
- **Quality Control:** Our audits of every split and 20 overlays per class checked by eye (Soccana, SoccerNet, GSR; 20 val overlays for MultiSports). GSR: players 20 of 20 and balls 18 of 20 correct, referees correct but 4 of 20 hard to see in blurred frames. Football spot-check of 34 images (5 per source, 4 from the smallest): Among 580 inspected boxes, no wrong class was found; one goalkeeper annotation was missing and three cases were unclear.

Quality Assessment (Data Audit)

- **Preprocessing Steps:** None on the raw data. Converted copies in YOLO format, lists of excluded images and the shared class scheme are kept separately. For inference, images are only resized to 640 or 1280 px.
- **Initial Audit Findings:** Football: 0 structural audit errors, 31 warnings and two separately identified label problems. Soccana: the test split leaks (135 test images have a twin in train), 81 boxes outside the image; 209 images excluded by the audit and 71 by the overlap check, and the group overlapping Football dropped. SoccerNet, GSR, MultiSports: 0 structural audit errors. SkillCorner: no missing frames; 1,534 player and 1,567 ball positions lie more than 5 m beyond the touchline or goal line. We use a 5 m margin so that normal play near the lines is not counted.
- **Quality Dimensions:** Completeness: every image has a label file, which may be empty; GSR pitch positions and SkillCorner positions are partly missing. Accuracy: labels mostly correct in our overlays; the Football spot-check found a missing annotation and unclear cases. Consistency: different class IDs and formats, unified in our converted copies. Timeliness: known match dates from 2012 to 2025; every version pinned by commit or revision. Zero-shot at 1280 px: person AP50 .83–.86 on SoccerNet and .95–.97 on real Football val, ball recall only .27–.32 on SoccerNet.

Legal, Ethical, and Governance

- **Sensitive Data:** Our source footage shows professional players, officials and spectators; faces are not blurred. We analyse players and teams without identifying individuals or linking tracks to names. SkillCorner includes names and birth dates, but we use player IDs only. We keep the datasets local and do not redistribute them.
- **Compliance:** We conduct this project exclusively for educational purposes at Lucerne University of Applied Sciences and Arts (HSLU). Our model detects players and the ball for football analysis, without facial recognition or individual identification. We treat the current pre-deployment research and development work under the EU AI Act research exclusion (Art. 2(8)); real-world testing and later deployment require their own assessment. We handle identifiable source footage and player metadata under applicable data-protection requirements (GDPR / Swiss FADP), using only the data needed for our analysis [4, 5, 17].
- **Consent & Licensing:** Our SoccerNet NDA permits both team members, Timo Tran and Johan Roth, to use the data for this HSLU project, provided we do not redistribute it. We cite the original sources. SoccerNet allows cited examples in research papers and presentations [12, 13]. Dataset-specific licences are summarised in Table 3.

Dataset	Licence	What the licence says	Consent and our use
Football Image Dataset	CC BY 4.0 (licence file of the dataset)	Use, share and adapt for any purpose if the author is credited. Covers the annotations; the frames are third-party broadcasts.	Consent of the people shown not mentioned. Coursework only, images not shared, author credited.
Soccana V1	No aggregate licence declared	Upstream licences apply. Five linked source pages declare CC BY 4.0; three source licences could not be verified. The card lists eight sources, while our local audit identifies seven groups. A complete source-to-file licence mapping is not established [1, 2, 8, 9, 19, 20].	Individual consent is not stated in the dataset card. No redistribution; training-use permission remains unresolved for unverified sources.
SoccerNet Tracking 2023	Research terms; no explicit licence on public card	SoccerNet permits model training for non-commercial research. Broadcast copyright remains with the rights holders. Cite the dataset; do not redistribute protected material [12, 13, 15].	Our NDA permits both of us to use the data for this HSLU educational project. We do not redistribute it. Individual consent is not stated in the public FAQ.
SoccerNet GSR 2025	GPL-3.0 tag on dataset card	The card declares GPL-3.0; its scope for third-party imagery is not clarified. Apply SoccerNet research restrictions to the footage; the tag is not treated as unrestricted broadcast rights [13, 14].	We use the data for our HSLU educational project under our SoccerNet NDA, without redistribution. We cite the paper. Individual consent is not stated in the dataset card.
SkillCorner Open Data	MIT in repository	Keep the copyright and permission notice when redistributing covered material. Credit SkillCorner. Repository licensing is distinct from personal-data permissions [11].	Contains names and birth dates; we use IDs only. Individual consent is not stated in the repository. IDs may remain linkable to players.
MultiSports	CC BY-NC 4.0	Attribution and non-commercial use required; indicate changes. The access page requires acceptance of the licence. Third-party footage may carry additional rights [7].	Access acceptance is recorded in our existing audit. Individual consent is not stated in the dataset card. Coursework only, paper cited.

Table 3: Licences and consent per dataset.

Dataset Structure and Schema

- **File Formats:** JPEG, YOLO `.txt`, COCO JSON, MOT `.txt` and `.ini`, GSR JSON, MP4 with a pickle, SkillCorner JSON, JSON lines and CSV.
- **Directory Layout:** Raw inputs include `data/raw/Football-Image-Dataset/` and `data/raw/soccana_v1/`. Converted evaluation data is under `data/processed/soccernet/`, `data/processed/gsr/` and `data/processed/sportsaction/`; audit outputs are kept separately under `outputs/`. Raw downloads are not edited.
- **Schema Summary:** Every image dataset is converted to YOLO: one label file per image, one line per box: `class x_center y_center width height`. SkillCorner: one tracking line per frame; events link to frames and player IDs.
- **Units & Encodings:** YOLO coordinates normalised to $[0, 1]$; COCO, MOT and GSR boxes in pixels; GSR and SkillCorner positions in metres; UTF-8 text.

Credits and References

We credit the original dataset providers and papers [1, 3, 6, 10, 11, 16]. YOLO models and software are provided by Ultralytics under their published licensing terms [18]. Source and terms pages were checked on 6 November 2026. The data counts and findings above come from our project audit.

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